WSInsight as a cloud-native pipeline for single-cell pathology inference on whole-slide images

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Abstract

Digital pathology has increasingly embraced deep learning for large-scale tissue analysis, but most workflows remain limited by local execution and data accessibility.To address this challenge, we developed WSInsight, an open-source, cloud-native pipeline designed to enable scalable single-cell pathology inference directly on whole-slide images. The integration of distributed processing, cloud storage access, and automated quality control streamlines the application and reuse of advanced deep learning methods, expanding opportunities for diagnostic, prognostic, and spatially resolved research in digital pathology.

Keywords: Whole-slide image, cloud-native, single-cell pathology, WSInfer, CellViT, HoVer-Net, StarDist and ResNet50.

Histopathology has long been the foundation of disease diagnosis, traditionally relying on the examination of tissue sections under high-power microscopy. The ongoing shift toward digital pathology—where glass slides are digitized as high-resolution whole-slide images (WSIs)—has transformed how tissue data are generated and analyzed [1]. Each WSI can exceed tens of gigabytes in size, presenting substantial challenges for data management, storage, and visualization. Simultaneously, the digital transition enables the application of advanced computational and deep learning techniques for diagnostic, prognostic, and spatially resolved analysis in pathology [2].

Deep learning has fundamentally reshaped digital pathology, offering unprecedented capability to characterize tissue architecture, cellular diversity, and molecular phenotypes from WSIs [3]. Numerous studies demonstrate its potential in detecting histologic patterns associated with immune activity, genetic alterations, and clinical outcomes [1]. However, much of this progress remains confined to isolated computational environments, where models are difficult to reproduce or deploy across institutions. Limited access to large-scale image datasets, inconsistencies in preprocessing workflows, and the lack of unified deployment frameworks continue to hinder reproducibility and large-scale validation [4]. Even when open-source models exist, their dependence on local storage and on-premise infrastructure restricts adoption by experimental and clinical researchers [5]. To fully realize the promise of deep learning in pathology, scalable and interoperable systems are required—systems capable of operating consistently across heterogeneous data sources, computational environments, and institutional settings.

WSInfer [4] addressed an important component of this need by providing a reproducible, modular framework for patch-level whole-slide inference. By standardizing model configuration, establishing a reusable Model Zoo, and enabling inference across diverse compute backends, WSInfer lowered technical barriers and supported broader adoption of deep learning models within the pathology community. Alongside related frameworks—such as TIA Toolbox [6], MONAI [7], SlideFlow [8], and PHARAOH [9]—WSInfer contributed to a growing ecosystem of open-source solutions for deploying trained models on WSIs. Yet these tools remain primarily focused on local or patch-based workflows and do not extend to cloud-native, single-cell–resolved analyses.

Here we introduce WSInsight, a cloud-native, end-to-end pipeline that unifies patch-level and single-cell inference for modern distributed computing environments. Building on WSInfer's robust patch-level paradigm, WSInsight expands the analytical scope to include nuclei segmentation, cell-type classification, and spatial profiling within a unified multi-resolution framework (Fig. 1). The WSInfer/WSInsight Model Zoo incorporates interoperable architectures that support end-to-end analysis across diverse tissue types and staining protocols, drawing from state-of-the-art models used in the field [6]. Direct access to cloud-hosted WSIs—including public repositories such as The Cancer Genome Atlas (TCGA) [10] and private S3-based platforms [11]—eliminates local storage requirements and enables scalable, multi-institutional workflows.

WSInsight enables fine-grained characterization of hematoxylin-and-eosin–stained tissues at single-cell resolution. The system quantifies cellular composition, spatial organization, and microenvironmental structure—capturing features such as immune infiltration, stromal distribution, and neoplastic boundaries [12]. High-resolution cellular maps can be integrated with spatial transcriptomic or proteomic data to relate morphological phenotypes to molecular states across tissue contexts [13], supporting multimodal studies of tumor ecosystems and other complex microenvironments directly from routine H&E slides.

To make these analyses broadly usable, WSInsight produces unified region- and cell-level outputs in standardized formats—GeoJSON for QuPath [14] and comma-separated value (CSV) for OMERO+ [15]. These interoperable outputs can be visualized and processed within existing digital pathology platforms, lowering technical barriers for end users and enabling reproducible large-scale workflows across institutions and computational environments.

A representative application of WSInsight is multi-scale integrative spatial analysis (Fig. 2). Patch-level models from the WSInfer Model Zoo—such as breast-tumor-resnet34.tcga-brca [16] and lymphnodes-tiatoolbox-resnet50.patchcamelyon [6, 17, 18]—identify tumor-enriched and lymphocyte-aggregated regions across the slide. These regional predictions are complemented by cell-level outputs from WSInsight-compatible models, such as CellViT/CellViT++ [19], resolves and classifies individual nuclei. When overlaid, the patch- and cell-resolved outputs provide a coherent view of the tumor microenvironment, enabling quantitative assessment of immune–tumor interactions, cellular composition, and spatial organization within pathologist-defined regions.

[Fig. 1 about here.]

[Fig. 2 about here.]

Methods

The overall WSInsight workflow is illustrated in Fig. 1, which outlines its three principal components: whole-slide image acquisition, model inference, and integrative visualization. In Step 1 (WSI acquisition), digital pathology slides can be accessed from either cloud-based repositories such as S3 or TCGA, or from local storage, allowing hybrid deployment across institutional infrastructures. In Step 2 (Inference), WSInsight interfaces with both the WSInfer/WSInsight Model Zoo and user-defined models to perform multi-model, integrative inference over the regions of interest (ROI) generated based on image quality control component, e.g., HistoQC [20]. Predictions from diverse deep learning architectures—such as segmentation, classification, or spatial inference models—are unified into cellular- and region-level probability maps. Finally, Step 3 (Visualization and integrative analysis) enables the outputs to be rendered directly within QuPath and OMERO+, where the cellular and regional predictions can be jointly explored. These interoperable layers support quantitative analysis of immune, epithelial, and neoplastic components, allowing users to derive interpretable spatial features that capture the cellular microenvironment and tissue heterogeneity. Collectively, this workflow highlights WSInsight's design philosophy: an integrated, cloud–local ecosystem that bridges model execution, visualization, and biological interpretation within a single, reproducible framework.

Inference Runtime

The WSInsight inference runtime performs deep learning–based analysis on WSIs at both the patch and single-cell levels, and is available as a command-line interface and a Python package. The runtime accepts three main inputs: a directory or cloud path containing WSIs, a trained model from the WSInfer/WSInsight Model Zoo or a user-provided model, and an output directory for results. Each WSI is processed through a standardized workflow: tissue regions are detected, subdivided into uniformly sized patches, and analyzed using patch-level classification models or single-cell segmentation and typing models, depending on the selected inference mode.

WSInsight supports a wide range of architectures—including patch-based classifiers, nuclei segmentation networks, and single-cell–level cell-type predictors—allowing multi-scale analysis within a unified framework. These interoperable outputs ensure that predictions—whether regional tissue classifications or cell-level annotations—can be visualized, aggregated, and quantitatively analyzed across datasets and computational environments.

Model Zoo

The WSInfer/WSInsight Model Zoo provides a curated collection of pretrained deep learning models for digital pathology, hosted on the Hugging Face Hub to support open access and reproducible research. Each model repository includes a model card documenting the model's purpose, architecture, and training data; pretrained weights serialized in TorchScript; and a standardized configuration file specifying input parameters and inference settings. This unified structure ensures consistent documentation, transparent reuse, and cross-platform portability. A public registry on Hugging Face maintains all verified entries in the zoo.

WSInsight is fully compatible with the WSInfer Model Zoo, allowing all existing WSInfer patch-level models to run directly within the WSInsight runtime without modification. This backward compatibility enables users to reuse, benchmark, and integrate prior patch-based models alongside new single-cell inference modules within a single, cloud-native pipeline.

Beyond patch-level inference, WSInsight extends the Model Zoo with single-cell–level H&E image analysis models, enabling nuclei segmentation, phenotype classification, and spatial context modeling. The current WSInsight single-cell collection centers on three families of architectures: (1) CellViT and HoVer-Net, both trained on the PanNuke dataset; and (2) StarDist–ResNet50, provided here as a pan-tissue model trained on pooled 10x Genomics Xenium single-cell datasets curated manually. These models support high-resolution analysis of tissue organization across tumor, immune, and stromal compartments. A compact comparison of these representative architectures is provided in Table 1.

The PanNuke dataset [21] comprises 19,000 H&E image tiles from multiple organs, each with expert per-nucleus annotations spanning five morphologic classes (neoplastic, inflammatory, connective, dead, epithelial). These annotations enable supervised training for robust nuclei segmentation and phenotype classification across diverse histologic contexts. Within WSInsight, PanNuke-trained CellViT and HoVer-Net models provide a standardized morphological vocabulary that can be reused across projects and tissue types, and their representative performance characteristics are summarized in Table 1.

To complement these morphology-based models, we additionally assembled a manually curated collection of 10x Genomics Xenium datasets spanning multiple tissue types, as listed in Table 2. Using QuST [22] to derive cell-level annotations from paired Xenium and H&E data, we constructed harmonized label vocabularies for each anatomical site, capturing cell types based on their morphologic phenotypes observed under H&E staining. The annotated cell types include adipocyte, basophil, chondrocyte, endothelial, eosinophil, epithelial, fibroblast_like, glial, hematologic_blast, lymphocyte, macrophage_like, malignant_epithelial, malignant_melanocytic, mast_cell, melanocyte, mesangial, neuron, neutrophil, osteoblast, osteocyte, perivascular, plasma_cell, and smooth_muscle. When cellular microenvironmental context is taken into account, additional phenotypes such as basaloid_progenitor, neuroendocrine, and pericyte may also be identified. These site-specific label sets are summarized Table 3.

All Xenium datasets in Table 2 are pooled into a single multi-tissue corpus, and the harmonized label schema in Table 3 is used to supervise a unified pan-tissue model. This strategy leverages the diversity of available tissues while maintaining a consistent output space for downstream spatial analysis. Together, the extended WSInsight Model Zoo spans patch-level, single-cell, and pan-tissue inference, enabling multi-scale integrative analysis across diverse tissue types and experimental modalities.

[Table 1 about here.]

[Table 2 about here.]

[Table 3 about here.]

Reporting summary

Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Code Availability

The WSInsight framework is open source under the Apache 2.0 license. The source code, container images, and Model Zoo configurations are available at https://github.com/huangch/wsinsight.

Author Contributions

C.H.H. conceived and led the WSInsight design and implementation. O.E.A. contributed model validation and dataset curation. D.F. oversaw oncology use cases and biological interpretation. All authors wrote and approved the manuscript.

Competing Interests

The authors are employees of Pfizer Inc. This work was conducted as part of their employment. The authors declare no other completing interests.

Funding

Not applicable.

Acknowledgements

We thank the WSInfer authors, digital pathology and computational biology communities for open-source resources, and acknowledge TCGA and 10x Genomics that enabled this study.

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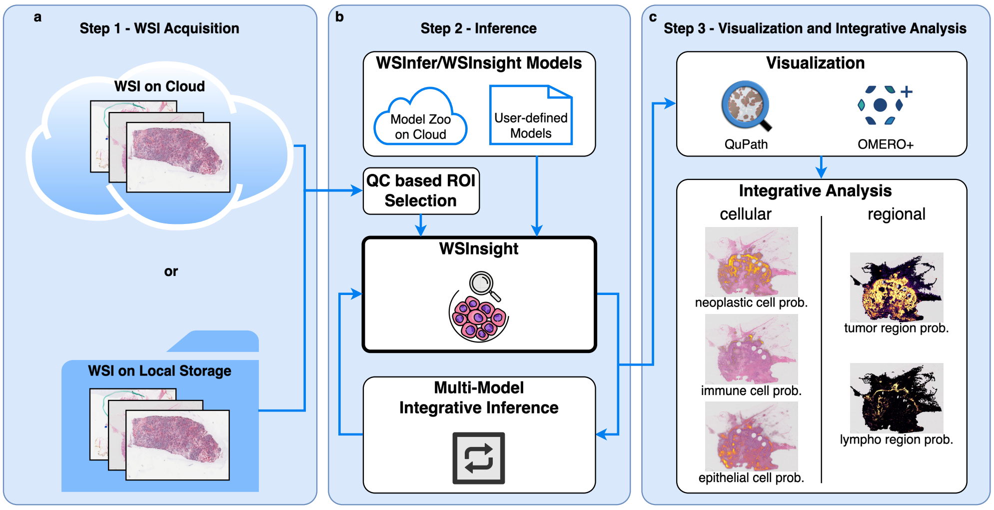
a
Step 1 - WSI Acquisition
WSI on Cloud
or
WSI on Local Storage
b
Step 2 - Inference
WSInfer/WSInsight Models
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User-defined Models
QC based ROI Selection
WSInsight
Multi-Model Integrative Inference
c
Step 3 - Visualization and Integrative Analysis
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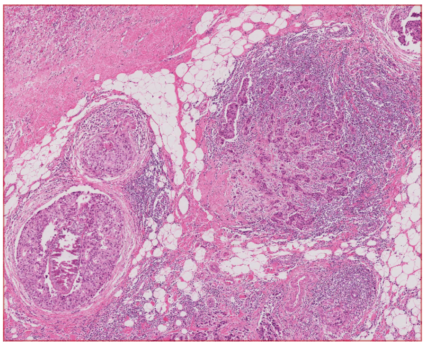
QuPath
OMERO+
Integrative Analysis
cellular
regional
neoplastic cell prob.
immune cell prob.
epithelial cell prob.
tumor region prob.
lympho region prob.



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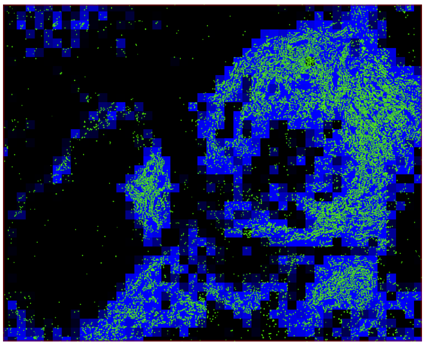

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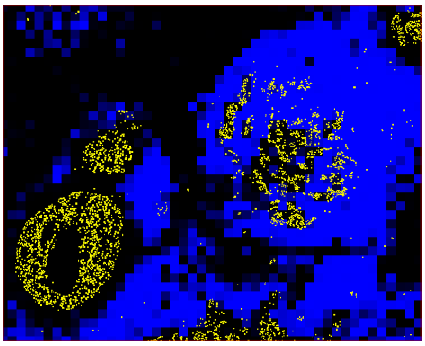
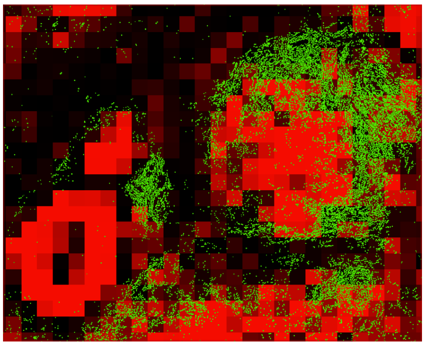
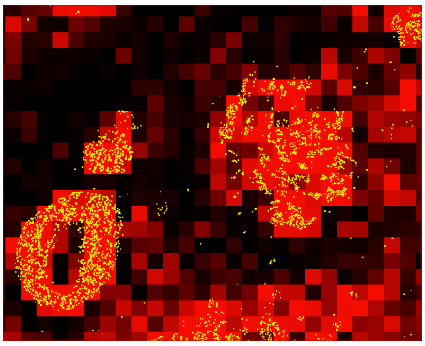


(c)



(d)



(e)

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Method	Architecture and Key Features	mPQ	bPQ
CellViT [19]	Vision Transformer encoder with U-Net style decoder; trained on multi-tissue datasets including PanNuke; supports multi-class nuclear instance segmentation and classification.	0.4980	0.6793
HoVer-Net [23]	CNN backbone (typically ResNet50) with dual-branch decoder predicting nuclear masks and horizontal/vertical (HoVer) distance maps for improved instance separation.	0.4629	0.6596
StarDist–ResNet50 [24, 25]	ResNet50 backbone paired with star-convex polygon representation predicting radial distances along fixed rays for nuclei delineation.	0.4796	0.6692

1. Human Bone and Bone Marrow Data with Custom Add-on Panel / Non-diseased Femur Bone
2. Human Bone and Bone Marrow Data with Custom Add-on Panel / Acute Lymphoid Leukemia Bone Marrow
3. Human Bone and Bone Marrow Data with Custom Add-on Panel / Non-diseased Bone Marrow
4. FFPE Human Brain Cancer Data with Human Immuno-Oncology Profiling Panel and Custom Add-on
5. FFPE Human Breast Cancer with 5K Human Pan Tissue and Pathways Panel plus 100 Custom Genes
6. FFPE Human Breast using the Entire Sample Area / Replicate 1
7. FFPE Human Breast using the Entire Sample Area / Replicate 2
8. FFPE Human Breast with Pre-designed Panel / Tissue sample 1
9. Xenium FFPE Human Breast with Custom Add-on Panel / Tissue sample 1 (IDC)
10. FFPE Human Breast with Custom Add-on Panel / Tissue sample 1
11. FFPE Human Breast with Pre-designed Panel / Tissue sample 2
12. FFPE Human Breast with Custom Add-on Panel / Tissue sample 2
13. FFPE Human Cervical Cancer with 5K Human Pan Tissue and Pathways Panel plus 100 Custom Genes
14. Human Colon Preview Data (Xenium Human Colon Gene Expression Panel) / Cancer / pre-designed + add-on panel
15. Human Colon Preview Data (Xenium Human Colon Gene Expression Panel) / Non-diseased / pre-designed panel
16. FFPE Human Colorectal Cancer Data with Human Immuno-Oncology Profiling Panel and Custom Add-on
17. Human Heart Data with Xenium Human Multi-Tissue and Cancer Panel
18. Human Kidney Preview Data (Xenium Human Multi-Tissue and Cancer Panel) / Kidney Cancer (PRCC)
19. Human Kidney Preview Data (Xenium Human Multi-Tissue and Cancer Panel) / Non-diseased Kidney
20. Xenium In Situ Gene and Protein Expression data for FFPE Human Renal Cell Carcinoma
21. Human Liver Data with Xenium Human Multi-Tissue and Cancer Panel / Liver cancer
22. Human Liver Data with Xenium Human Multi-Tissue and Cancer Panel / Non-diseased
23. Post-Xenium Technical Note: Xenium v1 and Xenium Prime 5K for FFPE Human Lung Cancer / Experiment 2/ Xenium Prime 5K
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26. Preview Data: FFPE Human Lung Cancer with Xenium Multimodal Cell Segmentation
27. Preview Data: FFPE Human Lymph Node with 5K Pan Tissue and Pathways Panel
28. FFPE Human Ovarian Cancer with 5K Human Pan Tissue and Pathways Panel plus 100 Custom Genes
29. FFPE Human Ovarian Cancer Data with Human Immuno-Oncology Profiling Panel and Custom Add-on
30. Pancreatic Cancer with Xenium Human Multi-Tissue and Cancer Panel
31. FFPE Human Pancreatic Ductal Adenocarcinoma Data with Human Immuno-Oncology Profiling Panel
32. FFPE Human Pancreas with Xenium Multimodal Cell Segmentation
33. Preview Data: FFPE Human Prostate Adenocarcinoma with 5K Human Pan Tissue and Pathways Panel
34. Preview Data: FFPE Human Skin Primary Dermal Melanoma with 5K Human Pan Tissue and Pathways Panel
35. Human Skin Preview Data (Xenium Human Skin Gene Expression Panel)
36. Human Skin Data with Xenium Human Multi-Tissue and Cancer Panel / Sample 1
37. Human Skin Data with Xenium Human Multi-Tissue and Cancer Panel / Sample 2
38. Human Skin Preview Data (Xenium Human Skin Gene Expression Panel with Custom Add-On)
39. Human Tonsil Data with Xenium Human Multi-Tissue and Cancer Panel / Follicular lymphoid hyperplasia
40. Human Tonsil Data with Xenium Human Multi-Tissue and Cancer Panel / Reactive follicular hyperplasia

Tissue Site	Labels	Training Dataset (see Table 2)
bone (incl. bone marrow)	adipocyte, chondrocyte, endothelial, fibroblast_like, hematologic_blast, lymphocyte, macrophage_like, malignant_epithelial, mast_cell, neutrophil, osteoblast, osteocyte, pericyte	1, 2, 3
brain	fibroblast_like, glial, lymphocyte, macrophage_like, neutrophil, pericyte	4
breast	adipocyte, endothelial, epithelial, fibroblast_like, lymphocyte, macrophage_like, malignant_epithelial, mast_cell, pericyte, plasma_cell	5, 6, 7, 8, 9, 10, 11, 12
cervix	basaloid_progenitor, endothelial, epithelial, fibroblast_like, lymphocyte, macrophage_like, malignant_epithelial, pericyte, smooth_muscle	13
colorectal	basaloid_progenitor, basophil, endothelial, eosinophil, epithelial, fibroblast_like, hematologic_blast, lymphocyte, macrophage_like, malignant_epithelial, mast_cell, neuron, neutrophil, pericyte, plasma_cell	14, 15, 16
heart	basophil, cardiomyocyte, endothelial, fibroblast_like, lymphocyte, macrophage_like	14, 15, 17
kidney	endothelial, epithelial, fibroblast_like, lymphocyte, macrophage_like, malignant_epithelial, mesangial, neutrophil, pericyte, perivascular, plasma_cell	18, 19, 20
liver	endothelial, epithelial, fibroblast_like, lymphocyte, macrophage_like, malignant_epithelial, pericyte, plasma_cell	21, 22
lung	basophil, endothelial, epithelial, fibroblast_like, hematologic_blast, lymphocyte, macrophage_like, malignant_epithelial, mast_cell, pericyte, plasma_cell, smooth_muscle	23, 24, 25, 26
lymph node	endothelial, fibroblast_like, hematologic_blast, lymphocyte, macrophage_like, mast_cell, pericyte, plasma_cell	27
pancreas	endothelial, epithelial, fibroblast_like, lymphocyte, macrophage_like, malignant_epithelial, mast_cell, neuroendocrine, neutrophil, pericyte, plasma_cell	30, 31, 32
prostate	endothelial, epithelial, fibroblast_like, lymphocyte, macrophage_like, malignant_epithelial, neutrophil, pericyte, plasma_cell, smooth_muscle	33
skin	basophil, endothelial, epithelial, fibroblast_like, lymphocyte, macrophage_like, malignant_epithelial, malignant_melanocytic, mast_cell, melanocyte, neuroendocrine, pericyte, plasma_cell	34, 35, 36, 37, 38
tonsil	basophil, endothelial, fibroblast_like, hematologic_blast, lymphocyte, macrophage_like, mast_cell, pericyte, plasma_cell	39, 40
pan-tissue	neoplastic, epithelial, inflamatory, connective, background	all of above